CGS616 - Assignment 1

Weather conditions and adaptive driving behaviour

Mohit Choudhary

Group Axiom

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1 Introduction

Driving is one of the most cognition-heavy tasks, which can be shaped by both internal and external factors. Both internal and external factors work hand-in-hand to determine the driver's behavior, decision making and hence affecting the consequences. In this paper, our primary focus will be weather conditions and how they affect driving behavior and these behaviors can lead to safety or accidents and other results.

For a beginner, driving is a conscious learning task which requires attention and attentive cognition but for an experienced driver, driving becomes subconsious tasks which no more requires much attention to details. But, as we change weather conditions from clear to foggy, visibility drops, adrenaline rush increases and visual cognitive load increases.

For rainy conditions, visibility drops due to water on the windshield which increases load on visual cognition. The sound of rain also affects driver's auditory cognition. The driver's state of mind shifts from subconscious to deliberate cognition and attention to details becomes important in order to take safety measures and avoid mishappening.

Snowy conditions not only affect visibility but also affect road conditions, temperature and traffic flows. The factors like friction, braking system efficiency, safe braking distance and lights to follow come into play.

We will follow ABC framework: Antecedent, Behavior, Consequence framework throughout this study. In our context antecedent refers to the weather conditions, behavior refers to driver's tendency to adhere to the rules and consequences are safety or mishappenings due to these behaviors.

We will use try to model traffic behavior using two behavioral states - following (compliance to traffic rules) vs not following (violation of traffic rules) and we will analyse how varying weather conditions affect these states and what factors cause transitioning between these states. We'll also try to comment on state probabilities under different conditions and safety outcomes in a particular condition.

2 Objective

The primary aim of this study is to analyse behavior of the driver in different weather conditions and comparing compliance vs violation of traffic rules behaviors. We will also try to determine the probability of existence of these states and corresponding consequences. The objectives of this study are as follows:

1. To analyze adaptive driving behavior in different weather conditions and their consequences using word clouds and frequency-based visualizations.

2. To compare compliance vs violation behaviour in these 4 conditions:-

    - Clear and sunny weather
    - Foggy weather
    - Rainy weather
    - Snow conditions

3. To identify these weather conditions as antecedents which control driving behavior

4. To explore the consequences according to driving behaviors

5. To do statistical analysis by calculating probabilities of different states

3 Methodology

The nature of this study is qualitative in nature. Instead of using large datasets and extracting information out of them, we have used qualitative text-based data for analysing adaptive driving behavior using ABC framework.

We have used 2 text based datasets for our analysis. We preferred text based datasets over numerical because we did not want this to come out as a data mining task but rather cognitive science project which also analyses cognition, perception and reasoning for the visible behaviors.

3.1 Survey form data

We floated a survey form which captured the age of the responder and provided them with 4 scenarios with 4 different weather conditions:

1. clear
2. foggy
3. rainy
4. snowy

For each of these scenarios we asked the audience about likelihood of compliance to rules and how it can lead to safety of accidents. This dataset not only provided us about how people perceive different driving conditions but also provided us with data for word cloud as we introduced subjective question regarding the same.

3.2 Data from News article headlines

To analyze media framing of traffic incidents, we utilized an automated data collection pipeline targeting the Google News RSS feed. Using a custom Python script with the feedparser library, we extracted recent headlines matching four distinct weather-related queries: 'car accident rain', 'car crash fog', 'traffic accident snow', and 'speeding accident clear weather'. This approach ensured the retrieval of high-relevance headlines from diverse publishers, eliminating manual selection bias and providing a structured dataset for textual analysis.

3.3 Word Cloud and frequency count

Generated word clouds for these 4 weather conditions to visualise the frequency of repeated terms for understanding the wholesome picture. We analysed this using both the datasets that we have. We combined news article headlines of a particular weather with survey responses about how people feel in that particular weather which gave us both a bigger picture overview from news and also minor considerations that humans take into account at individual levels.

Along with word clouds, a keyword frequency analysis was done to see how important certain words were in each of the four weather conditions (Clear, Rain, Fog, Snow). We chose keywords based on how they relate to accident attribution and cognitive state. Examples include speeding, skidding, visibility, and hydroplaning. Normalized frequency values were calculated to account for differences in the number of headlines collected per category, allowing for a meaningful comparison between datasets. A grouped bar graph was used to visualize shifts in keyword prominence across weather types. This comparative visualization supports a focused examination of how media framing shifts from internal attribution (e.g., driver error) to external attribution (e.g., environmental hazard) as weather conditions deteriorate.

3.4 The ABC Framework

This study is done using ABC (Antecedent-Behavior-Consequence) model:

- Antecedent: In general, antecedent refers to a trigger or a factor which leads to a certain behavior. In our context antecedents are the weather conditions: clear, rainy, foggy and snowy.
- Behavior: Behavior is the effort that a person makes to respond to the antecedent. In our context, we are trying to analyse two behaviors: compliance to rules and violation of rules.
- Consequence: It is the result following the behavior. In our case, safety and accidents are the two most common consequences that we are expecting.

We have decided to put forward this study in antecedent order i.e. we will analyse driving behavior from one weather condition to other weather condition. For each particular weather condition we will prepare the hypothesis and calculate the probability of driver's state and its consequences.

4 Results and Observations

4.1 Clear and Sunny Weather



Figure 1: word cloud for clear/sunny conditions



Figure 2: The ABC Model applied to Driving Behavior.

$$P(\text{Compliance}_{\text{clear}}) = \frac{60}{103} \approx 58.2\%$$

$$P(\text{Safety} \mid \text{Compliance}) = \frac{60}{60} = 100\%$$

$$P(\text{Safety}_{\text{total}}) = \frac{60}{103} \times 1 = 58.2\%$$

In clear weather conditions, the word cloud analysis highlights keywords such as 'Relaxed', 'Control', and 'Clear', indicating that the Antecedent (A) is characterized by low cognitive load and high environmental stability. This sense of security significantly influences the driver behavior (B) leading to the following of rules. According to the survey data traffic rule compliance is lowest in this weather condition with only 60/103 following and 43 not following the rules. The violation rate is much higher in this case which could be due to the clear weather. Drivers might perceive an illusion that they are safe and secure if the weather is clear i.e. they perceive low cost of violation. The data also shows that among rule-followers, 100% perceive themselves as safe.

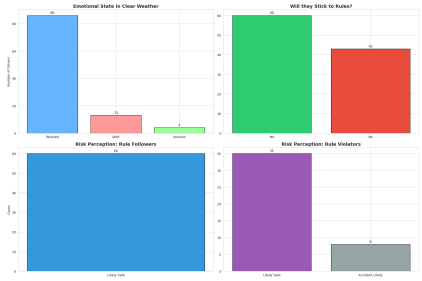


Figure 3: Survey Data for clear weather conditions

4.2 Rainy Weather

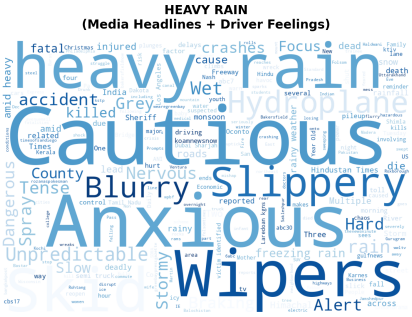


Figure 4: word cloud for rainy weather



Figure 5: The ABC Model applied to Driving Behavior.

$$P(\text{Compliance}_{\text{rain}}) = \frac{83}{103} \approx 80.6\%$$

$$P(\text{Safety} \mid \text{Compliance}) = \frac{76}{83} \approx 91.6\%$$

$$P(\text{Safety}_{\text{total}}) = \frac{83}{103} \times \frac{76}{83} = \frac{76}{103} \approx 73.8\%$$

As conditions shift to heavy rain, the word cloud transitions to sensory-descriptive terms like 'Anxious', 'Wipers', and 'Slippery'. With terms like cautious and anxious appearing in the word cloud, the cognition here might shift to deliberate and attentive rather than subconcious. The survey shows sharp rise in rule adherence, with 83 out of 103 (80.6%) respondents choosing to follow them. Unlike clear weather, where violation is "safe," the rain imposes a strict penalty. Among the 20 respondents who indicated they would not follow rules in these conditions, 100% believed they would likely face an accident. This shows that in adverse conditions like heavy rain drivers know that if they do not follow rules, it can lead to mishappening. Also, even if someone

follows rules, there are still 7/83 chances to face an accident due toh weather conditions.

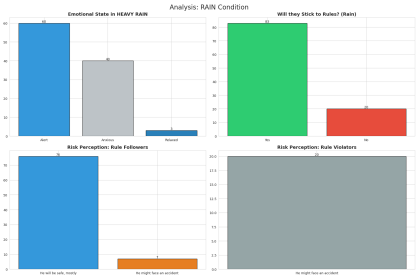


Figure 6: Survery Data for Rainy weather conditions

4.3 Foggy Weather

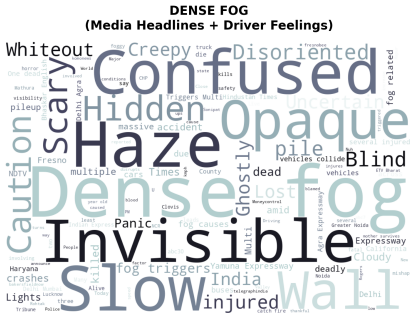


Figure 7: word cloud for foggy conditions

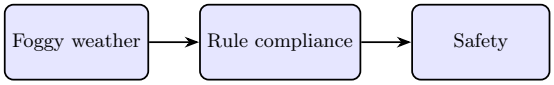


Figure 8: The ABC Model applied to Driving Behavior.

$$P(\text{Compliance}_{\text{fog}}) = \frac{93}{103} \approx 90.3\%$$

$$P(\text{Safety} \mid \text{Compliance}) = \frac{84}{93} \approx 90.3\%$$

$$P(\text{Safety}_{\text{total}}) = \frac{93}{103} \times \frac{84}{93} = \frac{84}{103} \approx 81.6\%$$

The word cloud for foggy weather is dominated by terms such as 'Haze', 'Invisible', and 'Confused', characterizing the Antecedent as severe sensory deprivation. This might make the driver extra-vigilant affecting behaviour (B).Our data shows that Rule Compliance jumps significantly to 93 out of 103 (90.3%). In this context, compliance is driven by fear rather than law. The Consequence (C) data reinforces this survival instinct: among the small group of 10 rule violators, 9 (90%) admitted they would likely face an accident. This indicates that the margin for error in fog is perceived as almost non-existent. Any deviation from strict compliance is viewed as almost an accident, which forces drivers to be attentive and comply with the rules.



Figure 9: Survery Data for Foggy weather conditions

4.4 Snow weather conditions

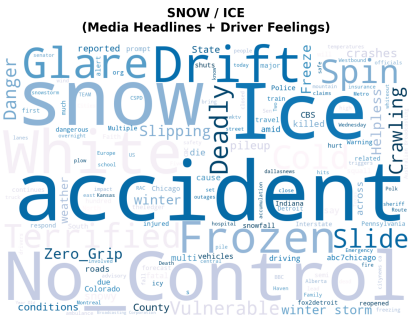


Figure 10: word cloud for snowy weather



Figure 11: The ABC Model applied to Driving Behavior.

$$P(\text{Compliance}_{\text{snow}}) = \frac{98}{103} \approx 95.1\%$$

$$P(\text{Safety} \mid \text{Compliance}) = \frac{83}{98} \approx 84.7\%$$

$$P(\text{Safety}_{\text{total}}) = \frac{98}{103} \times \frac{83}{98} \approx 80.6\%$$

Snow represents the most extreme Antecedent in our study, with word cloud descriptors like 'Slide', 'No Control', and 'Drift' indicating physical instability. This Antecedent dictates the most rigid Behavior (B) observed: 98 out of 103 (95.1%) respondents indicated they would comply with rules. This near-total compliance suggests that in snowy conditions, the choice to violate rules is effectively removed by the environment. The Consequence (C) data confirms this stark reality: for the tiny minority of 5 respondents who said they would not follow rules,

100% (5/5) acknowledged they would face an accident. There is zero perceived survivability for reckless behavior. Furthermore, even for those who do comply, the probability of safety drops to 84.7%, indicating that while compliance is necessary for survival, the environment is so hostile that even perfect behavior cannot guarantee safety.



Figure 12: Survey Data for Snowy weather conditions

5 Monte Carlo Simulation: The Evolution of Safety

While our survey data gives us a snapshot of a single moment, traffic safety is about consistency over time. To test this, we developed a Monte Carlo simulation to answer the question: *What happens when 1,000 drivers try to survive 20 consecutive trips in these conditions?*

How the Simulation Works?
We initialized $N = 1,000$ virtual agents for each weather type. For every trip ($t = 1$ to 20), the simulation runs two logical checks:

1. **Decision:** The driver decides to Comply or Violate based on the survey probabilities (e.g., 95% compliance in Snow).
2. **Consequence:** Based on that decision, a "dice roll" determines if they crash. If they crash, they are removed from the pool of safe drivers.

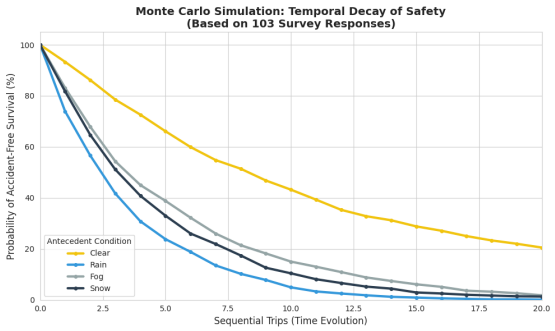


Figure 13: Monte Carlo simulation graph

Conclusion from the Graph: The results reveal a stark contrast. The **Clear Weather** line (Yellow) remains nearly flat, proving that "bad behavior" (rule violation) is sustainable because the risk per trip is low. Conversely, the **Snow** line (Dark Blue) drops precipitously. This mathematically demonstrates that in adverse weather, risk is cumulative; while a driver might survive one trip by luck, the probability of surviving 20 consecutive trips without an accident approaches zero, regardless of compliance.

6 Conclusion

In order to show that traffic rule compliance is a dynamic variable that is strictly controlled by environmental antecedents rather than a static trait, this study successfully operationalised the ABC (Antecedent-Behavior-Consequence) framework. Environmental stability and driver adherence to safety procedures are clearly inversely correlated, according to our analysis of 103 survey responses and media narratives.

The Transition of Compliance: The most significant finding is the dramatic transition in behavioral tendencies across the four conditions. In Clear and Sunny weather, the antecedent of high visibility and dry roads created a "safety buffer" that encouraged violation; only 58.2% (60/103) of drivers complied with rules, as the perceived cost of non-compliance was negligible. However, as the weather conditions became worse and worse, compliance became non-negotiable. The tendency to follow rules rose sharply to 80.6% in Rain, 90.3% in Fog, and peaked at 95.1% in Snow. This trend confirms that "Rule Compliance" is often a function of perceived risk rather than civic duty. In clear weather, compliance is a choice; in snow, it is a survival mechanism.

Cognitive Shift: From Autopilot and relaxed state to Hyper-Vigilance and anxious state, the word cloud data illustrates a fundamental cognitive shift. In clear weather, descriptors like 'control' and 'Music' indicate a reliance on System 1 (Automatic) cognition, where drivers operate on sub-conscious cognition styles, leading to complacency and Optimism Bias. Conversely, adverse conditions trigger a forced transition to System 2 (Deliberate) cognition. The driver becomes more focused and attentive as 'No Control' situation in fog and snow, representing a state of high cognitive load.